



5.1

Initial-Value Problems
for Ordinary Differential
Equations

The Elementary Theory of Initial-Value Problems

If a Lipschitz condition is proved, then the differential equation has a unique solution

Initial condition is present

$$\text{ex: } y' = 2y + 2 \quad y(0) = 2$$

Definition 5.1 A function $f(t, y)$ is said to satisfy a **Lipschitz condition** in the variable y on a set $D \subset \mathbb{R}^2$ if a constant $L > 0$ exists with

$$|f(t, y_1) - f(t, y_2)| \leq \underset{\substack{\text{Lipschitz constant} \\ \downarrow \\ y \text{ is changing}}}{L} |y_1 - y_2|,$$

whenever (t, y_1) and (t, y_2) are in D . The constant L is called a **Lipschitz constant** for f .

Example 1 Show that $f(t, y) = t|y|$ satisfies a Lipschitz condition on the interval $D = \{(t, y) \mid \underline{t \leq 2} \text{ and } -3 \leq y \leq 4\}$. ■

Solution For each pair of points (t, y_1) and (t, y_2) in D , we have

$$|f(t, y_1) - f(t, y_2)| = |t|y_1| - t|y_2|| = |t| ||y_1| - |y_2|| \leq 2|y_1 - y_2|.$$

Thus, f satisfies a Lipschitz condition on D in the variable y with Lipschitz constant 2. The smallest value possible for the Lipschitz constant for this problem is $L = 2$ because, for example,

$$|f(2, 1) - f(2, 0)| = |2 - 0| = 2|1 - 0|.$$

■

Suppose $f(t, y)$ is defined on a convex set $D \subset \mathbb{R}^2$. If a constant $L > 0$ exists with

$$\left| \frac{\partial f}{\partial y}(t, y) \right| \leq L, \quad \text{for all } (t, y) \in D, \quad (5.1)$$

$\frac{\partial}{\partial y}$ derivative with respect to one variable for a multi variable function
 δ

then f satisfies a Lipschitz condition on D in the variable y with Lipschitz constant L . ■

EXERCISE SET 5.1

- Use Theorem 5.4 to show that each of the following initial-value problems has a unique solution and find the solution.

a. $y' = y \cos t, \quad 0 \leq t \leq 1, \quad y(0) = 1.$

b. $y' = \frac{2}{t}y + t^2 e^t, \quad 1 \leq t \leq 2, \quad y(1) = 0.$

c. $y' = -\frac{2}{t}y + t^2 e^t, \quad 1 \leq t \leq 2, \quad y(1) = \sqrt{2}e.$

d. $y' = \frac{4t^3 y}{1 + t^4}, \quad 0 \leq t \leq 1, \quad y(0) = 1.$

1) a) $y' = y \cos t$, $0 \leq t \leq 1$, $y(0) = 1$ ^{Condition}

$$f(t, y) = y \cos t$$

$$f(t, y_1) = y_1 \cos t$$

$$f(t, y_2) = y_2 \cos t$$

$$|f(t, y_1) - f(t, y_2)| = |y_1 \cos t - y_2 \cos t| = |\cos t| |y_1 - y_2|$$

$$= 1 |y_1 - y_2|$$

$$\cos 0 = 1 \quad \cos 1 = 0.5$$

take the bigger value

$L = 1$, there for it has a unique solution

If L isn't a constant like ∞ , it won't have a unique solution

Another way:

$$\frac{\partial}{\partial y} (y \cos t)$$

$$\cos t(1)$$

$$= \cos t$$

$$|\frac{\partial}{\partial y} (y \cos t)| = \cos t$$

$$\leq 1$$

$$\cos 0 = 1 \leftarrow \text{bigger}$$

$$\cos 1 = 0.5$$

Finding solutions

$$dt \times \frac{dy}{dt} = y \cos t \times dt$$

$$\frac{dy}{y} = \frac{y \cos t}{y} dt \quad \frac{dy}{y} = \cos t dt$$

$$\int \frac{dy}{y} = \int \cos t dt \quad (\text{separable method (each variable on one side)})$$

$$\ln y = \sin t + C_1$$

$$y = e^{(\sin t + C_1)}$$

$$y = e^{\sin t} \cdot e^{C_1}$$

$$e^{C_1} = C$$

$$y(t) = C e^{\sin t}$$

$$y(0) = C e^{\sin 0} = 1$$

$$\text{given } C = 1$$

$$y(t) = e^{\sin t}$$

$$e^{\sin 0} = e^0 = 1$$

$$C \cdot 1 = 1$$

$$C = 1$$

$$\int \frac{1}{y} dy = \ln y$$

$$\int \cos t dt = \sin t + C$$

$$1) d) y' = \frac{4t^3 y}{1+t^4} \quad 0 \leq t \leq 1 \quad y(0) = 1$$

$$f(t, y) = \frac{4t^3 y}{1+t^4}$$

$$f(t, y) = \frac{4t^3}{1+t^4} y$$

$$f(t, y) = \frac{4t^3}{1+t^4} y$$

$$|f(t, y_1) - f(t, y_2)| = \left| \frac{4t^3}{1+t^4} y_1 - \frac{4t^3}{1+t^4} y_2 \right| = \left| \frac{4t^3}{1+t^4} \right| |y_1 - y_2|$$

$$\frac{4t^3}{1+t^4} \begin{cases} t=0 & \frac{4(0)^3}{1+0^4} = \frac{0}{1} = 0 \\ t=1 & \frac{4(1)^3}{1+1^4} = \frac{4}{2} = 2 \end{cases} \leftarrow \text{bigger so } L=2$$

therefore there is a unique solution

Note: $\ln a + \ln b = \ln(ab)$

$$C \ln a = \ln a^C$$

$$dx + \frac{dy}{y} = \frac{t^3}{1+t^4} dt$$

$$\int \frac{dy}{y} = \int \frac{t^3}{1+t^4} dt$$

$$u = 1+t^4$$

$$\int \frac{t^3}{1+t^4} dt \quad \frac{d}{dt}(1+t^4) = \frac{d}{dt}(u)$$

$$4t^3 = \frac{du}{dt}$$

$$\frac{4t^3 dt}{4} = \frac{du}{4}$$

$$= \int \frac{du}{4u}$$

$$= \frac{1}{4} \int \frac{du}{u}$$

$$= \frac{1}{4} \ln u$$

$$= \frac{1}{4} \ln(1+t^4)$$

$$\ln y = \frac{1}{4} \ln(1+t^4) + \ln C$$

$$\ln y = \ln(1+t^4)^{\frac{1}{4}} + \ln C$$

$$\ln a + \ln c = \ln ac$$

$$\ln y = \ln((1+t^4)^{\frac{1}{4}} C)$$

$$y = \sqrt[4]{1+t^4} C$$

$$y(0) = \sqrt[4]{1+0^4} + C = 1 \Rightarrow C = 0$$

$$y = (1+t^4)^{\frac{1}{4}}$$

$$y(1) = (1+1^4)^{\frac{1}{4}} = \sqrt[4]{1+1^4}$$

$$\sqrt[4]{1+0^4} + C = 1$$

$$1+C=1$$

$$C=0$$